PROBLEMS

Section 23.5: Spectra of Polyatomic Molecules

23.35 The infrared spectrum of HCN shows strong bands at 712.1 cm^{-1} and at 3312.0 cm^{-1}. There is a strong Raman band at 2089.0 cm^{-1}. There are weaker infrared bands at 1412.0 cm^{-1}, at 2116.7 cm^{-1}, at 2800.3 cm^{-1}, at 4004.5 cm^{-1}, at 5394 cm^{-1}, and at 6521.7 cm^{-1}. Identify these bands as fundamental, overtone, or combination bands, and give the shape of the molecule.

23.36 The N_2O molecule has three strong bands in its IR spectrum, at 588.8 cm^{-1}, at 1285.0 cm^{-1}, and at 2223.5 cm^{-1}. The band at 588.8 cm^{-1} has a Q branch. All three bands are fundamentals, and the molecule has been shown to be linear. Explain why CO_2, which is also linear, has only two fundamental IR bands whereas N_2O has three.

23.37 Using the frequencies in Problem 23.36, tell where to look for overtone and combination bands in the spectrum of N_2O.

23.38 The H_2S molecule has strong infrared bands at 1290 cm^{-1}, 2610.8 cm^{-1}, and 2684 cm^{-1}. There are weaker bands at 2422 cm^{-1}, 3789 cm^{-1}, and 5154 cm^{-1}. Assign these bands as fundamentals, overtones, or combination bands, and specify which normal mode corresponds to each fundamental.

23.39 The NO_2 molecule has strong infrared bands at 648 cm^{-1}, 1320 cm^{-1}, and 1621 cm^{-1}. There are weaker bands at 1373 cm^{-1}, 2220 cm^{-1}, and 2910 cm^{-1}. Assign these bands as fundamentals, overtones, or combination bands, and specify which normal mode corresponds to each fundamental.

23.40 **a.** Predict the shape of the CS_2 molecule.

b. Draw sketches depicting the vibrational modes of this molecule. Tell which will be seen in the Raman spectrum and which will be seen in the infrared spectrum.

c. Describe the microwave and rotational Raman spectra of this substance.

23.6 Fluorescence, Phosphorescence, and Photochemistry

In this section we discuss various processes that involve emission or absorption of photons. The material in this section is somewhat separate from spectroscopy, and the entire section can be skipped without loss of continuity.

Fluorescence and Phosphorescence

We discuss an example substance, benzophenone ($Ph_2C{=}O$, where Ph stands for the phenyl group, C_6H_5). Figure 23.15 shows schematically some low-lying electronic energy levels of benzophenone. The excited levels that are shown correspond to excitation of electrons in the carbonyl group. An excited level that is reached from the ground level by an $n \to \pi^*$ transition is labeled (n, π^*) and an excited level that is reached by a $\pi \to \pi^*$ transition is labeled (π, π^*).

There are two levels labeled (n, π^*), a singlet level ($S = 0$) and a triplet level ($S = 1$). There are two levels labeled (π, π^*), also a singlet and a triplet. The ground level is a singlet level. The singlet levels are labeled as S_0, S_1, S_2, and so on, in order of increasing energy. We reserve the subscript 0 for the ground level, so the triplet levels are labeled T_1, T_2, and so on. The selection rules allow absorption of a photon

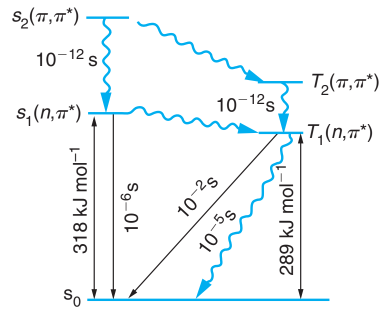

Figure 23.15 Some Energy Levels of the Benzophenone Molecule. The wavy arrows represent radiationless transitions, and the straight arrows represent emissions of photons. The times shown are relaxation times for the transitions. Data from D. L. Pavia, G. M. Lampman, and G. S. Kriz, Jr., *Introduction to Organic Laboratory Techniques*, 2nd ed., Saunders College Publishing, Philadelphia, 1982, p. 364.

to produce transitions only to excited singlet levels from the ground level. Transitions from the ground level to the two excited singlet levels give rise to two absorptions, one near 330 nm ($n \rightarrow \pi^*$) and one near 260 nm ($\pi \rightarrow \pi^*$).

If a molecule absorbs a photon to make a transition to an excited singlet level, the molecule can make the reverse transition and emit a photon of the same wavelength as the absorbed photon, but this is not the only thing that can happen. Because of the Franck–Condon principle, the molecule will probably be in an excited vibrational state after the upward transition so that the molecule can make a transition to a lower-energy vibrational level, losing some vibrational energy without changing the electronic state. This energy can be emitted as an infrared photon, or a *radiationless transition* can occur, in which case the vibrational energy lost by the molecule is transferred to other vibrational modes in the molecule or to rotation or translation of the molecule or to other molecules.

Once the molecule is in a lower vibrational level of the excited singlet level, it can emit a photon and return to the ground electronic level. Such a radiative transition to the ground level from an excited level with the same value of S is called *fluorescence*. Since vibrational energy was lost, the emitted light will have a longer wavelength than the absorbed light. Many common objects, including human teeth, certain minerals, and blacklight posters, can fluoresce, emitting visible light after absorbing ultraviolet light.

Another possibility is that the molecule might make a radiationless transition to the ground level or to a lower-energy electronic level with the same value of S. Such a radiationless transition is called an *internal conversion*. In our example of a carbonyl compound, an internal conversion could occur from the singlet (π, π^*) level to the singlet (n, π^*) level, followed by fluorescence to the ground level. Still another possibility is a radiationless transition to an electronic level with a different value of S, called an *intersystem crossing*. To each of the excited singlet levels in Figure 23.15, there corresponds a triplet level with the same electron configuration that can be reached by an intersystem crossing. After an intersystem crossing, the molecule might make a radiative transition to the ground state. This process is called *phosphorescence*. Since this process is forbidden by our approximate selection rules, a typical mean time for phosphorescence is longer than for fluorescence (typically 1 ms to 10 s). In Figure 23.15, the approximate values of mean transition times are indicated near each arrow.

Photochemistry

A molecule in an excited electronic state can possibly undergo a chemical reaction that is inaccessible in the ground level. If the excited state was reached directly or indirectly by absorption of radiation, the reaction is a *photochemical reaction*. Most photochemical reactions are governed by the *Stark–Einstein law of photochemistry*, which states that absorption of one photon causes the reaction of one molecule. However, the number of molecules that react is not necessarily equal to the number of photons absorbed. Some of the excited molecules might undergo internal conversion, intersystem crossing, fluorescence, or phosphorescence processes leading to unreactive states and therefore not react chemically. A *chain reaction* might occur in which the reaction of one molecule can lead to the reaction of other molecules without absorption of additional radiation.

The *quantum yield* of a photochemical reaction, Φ, is defined by

$$\Phi = \frac{\text{number of molecules reacted}}{\text{number of photons absorbed}} \tag{23.6-1}$$

Equation (23.6-1) can also be stated in terms of moles of reactant and moles of photons. One mole of photons is called an *einstein*, so that

$$\Phi = \frac{\text{amount reacted in moles}}{\text{amount of photons absorbed in einsteins}} \tag{23.6-2}$$

In a chain reaction, Φ can exceed unity, but in a nonchain reaction, $\Phi \leq 1$.

Upon radiation with ultraviolet light of 300 nm to 350 nm wavelength, benzophenone undergoes a reaction with 2-propanol to form benzpinacol and acetone.[21]

$$\mathrm{Ph_2C{=}O} + \mathrm{(CH_3)_2HCOH} \xrightarrow{h\nu} \mathrm{Ph_2}\underset{}{\overset{\mathrm{HO}\;|}{\mathrm{C}}} - \overset{\mathrm{OH}\;|}{\mathrm{C}}\mathrm{Ph_2} + \mathrm{(CH_3)_2C{=}O} \tag{23.6-3}$$

Because radiation of 300 nm wavelength has photons of insufficient energy to reach the $S_2(\pi, \pi^*)$ level, and because the radiative transition to a triplet level is forbidden, the first step in the mechanism for this reaction must be absorption of radiation to excite the benzophenone to the $S_1(n, \pi^*)$ level:

$$\mathrm{Ph_2C{=}O} + h\nu \longrightarrow \mathrm{Ph_2C}\dot{=}\dot{\mathrm{O}}\,(\mathrm{S_1}) \tag{23.6-4}$$

where the electron remaining in the nonbonding orbital is represented by a dot over the oxygen atom and the electron that has made the transition to the antibonding π orbital is represented by a dot over the double bond (which is now a bond with order 3/2).

The next step in the mechanism is an intersystem crossing to the T_1 level:

$$\mathrm{Ph_2C}\dot{=}\dot{\mathrm{O}}(\mathrm{S_1}) \xrightarrow{\text{isc}} \mathrm{Ph_2C}\dot{=}\dot{\mathrm{O}}(\mathrm{T_1}) \tag{23.6-5}$$

This step is followed by the abstraction of a hydrogen atom (complete with one electron) from a 2-propanol molecule by a benzophenone molecule in the T_1 level, forming a

[21] D. L. Pavia, G. M. Lampman, and G. S. Kriz, Jr., *Introduction to Organic Laboratory Techniques*, 2nd ed., Saunders, Philadelphia, 1982, p. 362ff.

free radical:

$$\mathrm{Ph_2C{=}\dot{O}(T_1) + H{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}{-}OH \longrightarrow Ph_2\dot{C}{-}OH + \underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{\dot{C}}}}}{-}OH} \tag{23.6-6}$$

The next step is the abstraction of a second hydrogen atom from the 2-propanol molecule by another excited benzophenone molecule, forming another radical and an acetone molecule (one of the final products):

$$\mathrm{Ph_2C{=}\dot{O}(T_1) + HO{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{\dot{C}}}}} \longrightarrow Ph_2\dot{C}{-}OH + O{=}C(CH_3)_2} \tag{23.6-7}$$

The final step is combination of two radicals:

$$\mathrm{2Ph_2\dot{C}{-}OH \longrightarrow Ph{-}\underset{\displaystyle Ph}{\overset{\displaystyle HO}{\underset{|}{\overset{|}{C}}}}{-}\underset{\displaystyle Ph}{\overset{\displaystyle OH}{\underset{|}{\overset{|}{C}}}}{-}Ph} \tag{23.6-8}$$

The photochemical reaction can be carried out by use of an ultraviolet lamp, but sunlight contains enough ultraviolet light to produce a significant amount of product in a few days. The reaction will proceed in a borosilicate glass flask. Borosilicate glass blocks almost all radiation of wavelength shorter than 300 nm, so the radiation that causes the reaction to proceed must have wavelengths longer than 300 nm.

Exercise 23.9

Calculate the energy per photon and per einstein for radiation of wavelength equal to 300 nm.

If naphthalene is placed in the reaction mixture, no reaction takes place. We say that the reaction is *quenched*. The explanation is that a rapid intermolecular energy transfer from an excited benzophenone molecule to a naphthalene molecule returns the benzophenone molecule to its ground level, before it can react chemically. Naphthalene has a singlet ground level a singlet (π, π^*) level 4.1 eV above the ground level, and a triplet (π, π^*) level 2.7 eV above the ground level. A well-obeyed selection rule requires that in an intermolecular energy transfer the sum of the two electron spin quantum numbers remains constant. This means that if the benzophenone molecule makes a transition from a triplet excited level to a singlet ground level, the naphthalene molecule must make a transition from its singlet ground level to a triplet excited level. If the benzophenone molecule makes a transition from a singlet excited level the naphthalene molecule must make a transition to a singlet excited level.

Since the (π, π^*) singlet excited level of naphthalene lies higher than the (n, π^*) singlet excited level of benzophenone by 0.8 eV, this level cannot be reached by energy transfer from a benzophenone molecule in its (n, π^*) level. However, the (π, π^*) triplet excited level of the naphthalene molecule lies lower than the (n, π^*) triplet level of benzophenone, so that this level can be reached by energy transfer from a benzophenone molecule. The fact that the naphthalene quenches the reaction shows that the triplet (n, π^*) level of benzophenone must be the reactive level.

Vision

Photochemical reactions are involved in vision in vertebrates.[22] There are two kinds of light-sensitive cells in the retina, called *rods* and *cones*. The rods provide for vision in dim light but do not provide color vision. Color vision is provided by three varieties of cone cells that are sensitive to red, green, and blue light respectively, and these require greater illumination than do the rod cells.

In the rod cells there is a protein called *rhodopsin*, which consists of a protein called *opsin* bonded to a polyene called *retinal*. Retinal is related to retinol, which is known as vitamin A and which is depicted in Figure 23.16a. Retinal occurs in the eye as the all-*trans* isomer and as the 11-*cis* isomer. The structural formulas of these isomers are shown in Figures 23.16b and 23.16c. The 11-*cis* isomer attaches to the free NH_2 group of a lysine residue in opsin, forming a Schiff base, as shown in Figure 23.16d. The all-*trans* isomer does not bond to the opsin. Rhodopsin has a broad absorption ranging from 400 nm to 600 nm, with maximum absorption around 500 nm. The corresponding absorption band of 11-*cis*-retinal is centered at 380 nm, in the ultraviolet. Each variety of cone cell has one of three proteins that are similar to rhodopsin, but which absorbs light only in either the red, green, or blue wavelength region.

Exercise 23.10

a. Using the structural formulas in Figure 23.16 and the free-electron molecular orbital (particle-in-a-box) model for a conjugated polyene, explain why the absorption maximum of the Schiff base form of rhodopsin is at longer wavelength than that of 11-*cis*-retinal.

b. Using the free-electron model, calculate the wavelength of maximum absorbance for 11-*cis*-retinal and for rhodopsin, taking an average bond length of 1.39×10^{-10} m and adding one bond length to each end of the conjugated system of bonds. Remember to count the pi electrons and assign two to each space orbital according to the Aufbau principle.

The accepted mechanism of the photochemical process in rod cells is as follows. First, the rhodopsin absorbs a photon, raising it to an excited state in which a 90° rotation has occurred about the double bond between carbons 11 and 12 of the retinal, making the molecule intermediate in shape between the all-*trans* isomer and the 11-*cis* isomer. Some of these molecules (about two-thirds) convert into the all-*trans* form, called bathorhodopsin. The retinal is still attached to the opsin, and this protein now undergoes a sequence of transformations, producing a series of identifiable proteins called lumirhodopsin, metarhodopsin I, and metarhodopsin II. A signal is apparently sent into a fiber of the optic nerve when metarhodopsin II undergoes a conformational change. Over a period of several minutes, the metarhodopsin II dissociates into opsin and free all-*trans*-retinal, which can be converted to the 11-*cis* form and attached again to opsin. The length of time required for this process is related to the time required for the eye to become dark adapted, but is much too slow to be involved in the actual process of vision.

[22] G. L. Zubay, *Biochemistry*, Addison-Wesley, Reading, MA, 1983, p. 409ff.

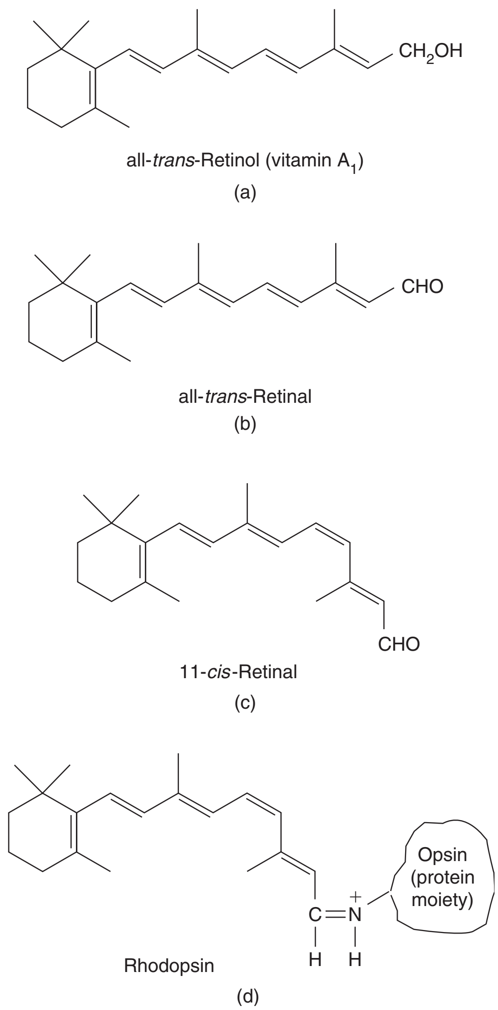

Figure 23.16 The Structures of Retinal and Rhodopsin. (a) All-*trans*-retinol (vitamin A) (b) All-*trans*-retinal. (c) 11-*cis*-retinal. (d) Rhodopsin.

PROBLEMS

Section 23.6: Fluorescence, Phosphorescence, and Photochemistry

23.41 Find the frequency and wavelength of light given off by the benzophenone molecule in (a) fluorescence from the S_1 state, and in (b), phosphorescence from the T_1 state.

23.42 In a photochemical reaction, 0.00100 watt of radiant energy of wavelength 254 nm is incident on the reaction vessel, and 61.2% of this energy is absorbed. If 2.45 mole of the absorbing reactant reacts to form products in 30.0 seconds, find the quantum yield.

23.43 Which of the following substances could quench the photochemical reaction of benzophenone discussed in Section 23.6?

Substance	Energy of First Triplet State/eV
Biphenyl	2.9
Toluene	3.6
Benzene	3.7
9,10-Diphenylanthracene	1.8
trans-1,3-Pentadiene	2.6

23.7 Raman Spectroscopy

Raman spectroscopy was invented by Chandrasekhara Venkata Raman, 1888–1970, an Indian physicist who received the 1930 Nobel Prize in physics for this work.

In Raman spectroscopy, radiation is inelastically scattered by the sample substance, either giving energy to or accepting energy from the substance. You can think of the process as consisting of absorption of photons of one frequency accompanied by emission of photons of a different frequency, with the difference in photon energy being absorbed or given off by the molecule. Figure 23.17 shows a schematic diagram of a Raman spectrometer. The scattered radiation is observed at right angles to the incident beam. It is generally much weaker than the incident beam, and lasers are now used to provide intense incident beams.

The difference in the photon energies of the incident and scattered radiation must equal the energy difference between two energy levels of the sample molecules. We denote the frequency of the incident radiation by ν and the frequency of the scattered radiation by ν'. If the radiation loses energy to the molecules,

$$h\nu - h\nu' = E_{\text{upper}} - E_{\text{lower}} \tag{23.7-1}$$

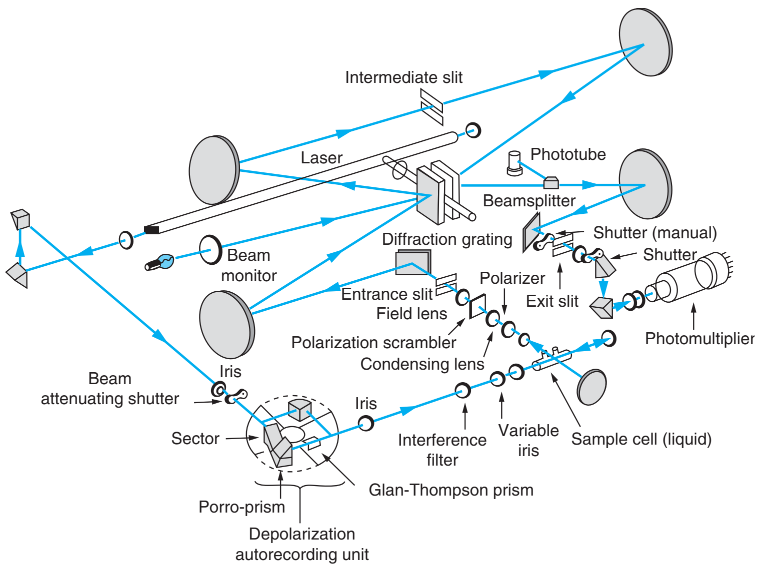

Figure 23.17 Schematic Diagram of a Laser Raman Spectrophotometer. In the diagram, there are two distinct beams: the incident beam, and the scattered beam at right angles to the incident beam. Courtesy of Jeol, Ltd.